Summary Notes: Cryto

• Photosynthesis - Lecture Slides (Secondary Level)**

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SLIDE 1 - TITLE

• Photosynthesis**

• How plants turn sunlight into food and oxygen*

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SLIDE 2 - LEARNING OBJECTIVES

By the end of this lesson you will be able to:

1. **Identify** the main parts of a chloroplast (outer membrane, inner membrane, thylakoids, grana, stroma).

2. **Explain** the light-dependent reactions that make ATP + NADPH and release O■.

3. **Describe** the Calvin-Benson (C■) cycle that fixes CO■ into a sugar (G3P).

4. **Compare** C■, C■ and CAM photosynthetic pathways and **state** why each evolved.

• Speaker note:* Emphasise that these objectives move from "what it looks like" → "what it does" → "why it matters".

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SLIDE 3 - WHY PHOTOSYNTHESIS MATTERS

• Converts solar energy into chemical energy stored in glucose.

• Occurs in **green plants, many algae, and some bacteria** - the base of every food web.

• The O■ we breathe comes from **splitting water (photolysis)** during the light reactions.

• Overall equation (simplified):

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6\text{CO}*2 + 6*text{H}*2\text{O} + \text{light} \;\rightarrow\; \text{C}*6*text{H}*{12}\text{O}*6 + 6*text{O}*2

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• Links physics (light), chemistry (redox), and biology (growth).

• Speaker note:* Use a short animation that shows sunlight → plant → food & oxygen.

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SLIDE 4 - CHLOROPLAST STRUCTURE & PIGMENTS

• **Double membrane** encloses the organelle.

• Inside: a fluid-filled **stroma** (site of the Calvin cycle).

• **Thylakoid membranes** are stacked into **grana**; each thylakoid is a flattened sac.

• **Pigments**

• **Chlorophyll a** - primary light-absorber (blue-red).

• **Chlorophyll b** & **carotenoids** - accessory pigments that capture additional wavelengths.

• Pigments act like solar panels: they **absorb photons**, become **excited**, and pass energy to **reaction-center proteins** in **photosystems**.

• Speaker note:* Point to a labeled diagram; highlight where light is captured vs. where sugar is made.

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SLIDE 5 - LIGHT-DEPENDENT (PHOTO-CHEMICAL) REACTIONS

1. **Location:** thylakoid **membranes**.

2. **Photolysis:** Light energy splits water → **O■ + H■ + e■**.

3. **Photosystem II (PSII)** captures light, boosts electrons, and passes them to **plastoquinone (PQ)**.

4. Electrons travel through **cytochrome b■f** → **plastocyanin (PC)** → **Photosystem I (PSI)**.

5. While moving, **protons (H■)** are pumped into the thylakoid lumen, creating a **proton gradient**.

6. **PSI** re-excites the electrons; **ferredoxin-NADP■ reductase** uses them to reduce **NADP■** → **NADPH** (high-energy carrier).

7. The **proton gradient** drives **ATP synthase** to make **ATP** from ADP + P■ (photophosphorylation).

• Key terms defined inline:*

• **Photolysis** - splitting of water by light.

• **Electron transport chain** - series of proteins that move electrons and pump protons.

• **Chemiosmosis** - using a proton gradient to make ATP.

• Speaker note:* Compare this chain to the mitochondrial chain to reinforce the concept of chemiosmosis.

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SLIDE 6 - CALVIN-BENSON (C■) CYCLE - CARBON FIXATION

• **Location:** stroma (fluid surrounding thylakoids).

• **Inputs:** 3 ATP + 2 NADPH per CO■ (produced in light reactions).

• Three phases**

1. **Carbon fixation** - **Rubisco** attaches CO■ to **ribulose-1,5-bisphosphate (RuBP)** → unstable 6-C → splits into two **3-phosphoglycerate (3-PGA)** molecules.

2. **Reduction** - ATP converts 3-PGA → **1,3-bisphosphoglycerate**; NADPH reduces it → **glyceraldehyde-3-phosphate (G3P)**.

3. **Regeneration** - Most G3P is recycled, using additional ATP, to reform RuBP so the cycle can continue.

• **Outcome:** For every 6 CO■ fixed, **1 glucose** (or 2 G3P) is produced, costing **18 ATP** and **12 NADPH**.

• Speaker note:* Use a "investment-payoff" graphic: energy spent first → sugar earned later.

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SLIDE 7 - C■ vs C■ vs CAM PATHWAYS

| Feature | **C■ (Calvin-Benson)** | **C■** | **CAM** |

|---|---|---|---|

| **Primary CO■-fixing enzyme** | Rubisco (in mesophyll) | **PEP carboxylase** (captures CO■ in mesophyll) → C■ acids → transported to bundle-sheath cells where Rubisco works | Same as C■ but **temporal separation**: night (CO■ uptake) → day (photosynthesis) |

| **Water use** | Higher (stomata stay open) | Lower (stomata close more) | Very low (stomata open at night) |

| **Photorespiration** | High (Rubisco also reacts with O■) | Low (CO■ concentrated around Rubisco) | Very low |

| **Typical habitats** | Temperate, moderate climate | Hot, sunny, high-light environments (e.g., tropical grasses) | Arid or semi-arid (e.g., cacti, pineapple) |

| **Evolutionary driver** | Baseline pathway | **Heat & drought** → need to conserve water & limit photorespiration | **Extreme drought** → store CO■ at night |

• Why they evolved:* To **increase efficiency** when Rubisco's oxygenase activity becomes a problem (high temperature, low CO■) and to **save water**.

• Speaker note:* Show a side-by-side diagram of leaf anatomy for C■ vs C■ and a timeline diagram for CAM.

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SLIDE 8 - REAL-WORLD APPLICATIONS

• **Food production:** C■ crops (maize, sugarcane) yield more in hot, dry regions.

• **Climate mitigation:** Forests & oceans act as massive **carbon sinks** via photosynthesis.

• **Synthetic biology:** Engineers are redesigning Rubisco or adding C■ traits to C■ crops to boost yields.

• **Solar-fuel research:** The light-dependent electron chain inspires artificial systems that split water or make fuels.

• **Agricultural biotech:** Gene-editing to reduce photorespiration or improve water-use efficiency.

• Speaker note:* Ask students to name a local crop and discuss which pathway it uses.

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SLIDE 9 - FORMATIVE CHECK & CRITICAL THINKING

• Quick-Check Questions (think-pair-share)**

1. What molecule is the **direct source of O■** released during photosynthesis?

2. Which pigment absorbs **green light the least**?

3. In C■ plants, where does the **Calvin cycle** actually occur?

• Discussion Prompts**

• *How would a rise in atmospheric CO■ affect photorespiration in C■ plants?*

• *If you could add one feature from C■ or CAM to a C■ crop, which would you choose and why?*

• Mini-Experiment Idea**

• Test the effect of light colour on photolysis:*

• Isolate chloroplasts, expose them to red, blue, and green LEDs, measure O■ evolution with a gas-sensor.

• Speaker note:* Circulate while groups discuss; reconvene to share key ideas.

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SLIDE 10 - SUMMARY & CONCLUSION

• **Photosynthesis** = sunlight → **ATP + NADPH** (light reactions) → **glucose** (Calvin cycle).

• **Chloroplast architecture** (thylakoids, grana, stroma) is essential for capturing light and running chemistry.

• **C■ & CAM** are clever adaptations that reduce water loss and photorespiration.

• Understanding these processes helps us improve crops, design renewable energy, and combat climate change.

• Design Challenge:* If you could redesign one component of the photosynthetic system to maximize food production, what would it be and why?*

• Further Reading**

- Textbook chapter on "Photosynthetic Adaptations" (e.g., *Campbell Biology*).
- Review article: "Engineering C■ traits into C■ crops" - *Nature Plants* (2023).
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